

MUTHUKUMARAN YOGEESWARAN

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EXPERIENCE

Research Student, NUS Immersive Reality Lab Aug 2025 - Present

- Designing an AI social companion for eldercare, focusing on emotional wellbeing, engagement, and human-robot interaction, in collaboration with NUS Nursing.
- Prototyping expressive conversational behaviours for a small animatronic companion robot with motor-driven movements.

Robot Operator, Cortex AI Nov 2025 - Jan 2026

- Teleoperated robotic systems to execute precision tasks, collecting egocentric vision data and collaborating with engineering teams to improve system accuracy.

Research Internship, The University of Osaka May 2025 - Aug 2025

- Developed a real-time human detection and localisation system using YOLOv12 for a mobile Teleco robot in social robotics applications.
- Integrated external camera perception with SLAM-based mapping; system deployed at the Yoshikawa Lab pavilion at Osaka Expo 2025.

Software Team Member, NUS RoboMaster Aug 2023 - Dec 2024

- Collaborated within a multidisciplinary team of 4 to design, simulate and build a 2 legged balancing robot for participation in the Da-Jiang Innovations (DJI) RoboMaster Championships.
- Prototyped an aim-bot utilising computer vision, for integration with robots to gain a competitive edge in upcoming competitions.

Research Internship, ASTAR May 2024 - Aug 2024

- Implemented reinforcement learning in MuJoCo to train a 6-DOF Inspire dexterous hand for object reallocation tasks leveraging human video demonstrations.

PROJECTS

Software Lead, CDE4301 Innovation & Design Capstone Aug 2025 - Present

- Leading end-to-end robotics software development for an intelligent boxing robot, implementing ROS-based system architecture and integrating perception, action recognition, and feedback modules.
- Developing neural network-based models for pose estimation and action recognition, and integrating fine-tuned LLMs to generate real-time, domain-specific boxing feedback.
- Designing and curating a custom RGB-D action dataset using an Intel RealSense D435i, including developing an in-house video annotation GUI to support supervised temporal labelling and model evaluation.

Project Co-Lead, NUS Orbital (Software Development Project) May 2023 - Aug 2023

- Developed a gamified lifestyle mobile application in a team of 2, employing React Native and Firebase technologies.
- Spearheaded design process using Figma for mockup creation and managed version control with Git and GitHub.

TECHNICAL SKILLS AND PROFESSIONAL CERTIFICATIONS

- Programming Languages: Python, C++, C, MATLAB, JavaScript, Assembly
- Machine Learning & AI: Computer Vision, Neural Networks, Action Recognition (Pose / Temporal Models), Reinforcement Learning
- Robotics & Systems: ROS2, SLAM, Robot Perception, Mobile Robotics, STM32
- Simulation & Control: MuJoCo, COMSOL Multiphysics, SolidWorks Simulation, PID/LQR Control, Simulink
- CAD & Design: SolidWorks (CSWA – Mechanical Design & Additive Manufacturing)
- Tools & Platforms: Git, GitHub, Docker, Conda, Linux, React Native, Firebase

EDUCATION

National University of Singapore
Bachelor of Engineering

Aug 2022 - Jul 2026

- Major in Engineering Science with a specialisation in Computational Engineering Science
- Minors in Innovation & Design Programme and Computing (Design and Engineering)
- Focus areas: Robotics, Artificial Intelligence, Human–Robot Interaction

RELEVANT COURSEWORK

- Mobile Robotics & Perception (ROS)
- Machine Intelligence & Neural Networks
- Signals and Systems
- Data Structures & Algorithms
- Sensor Systems Electronics

EXTRA CURRICULAR ACTIVITIES

NUS Student Ambassador, National University of Singapore

Jan 2023 - Present

- Represent NUS as a Student Ambassador, supporting communications, outreach, and official events under the Office of University Communications.

President, NUS Engineering Science Student Wing

Aug 2023 - Aug 2024

- Led a team of 40 members to strategize, plan, and coordinate a variety of academic and recreational events for the student body.
- Planned and hosted a prototyping competition (MakerMania23) where roughly ~40 teams of 4-5 students from various Junior Colleges, Polytechnics and Universities competed in.